Tutorials on Areas, Volumes, Line Integral, Surface Integrals, Green's Theorem, Gauss Divergence Theorem, Stoke's Theorem

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(1) Calculate $r^2 dr d\theta$ over the area included between the circles $r=2\sin\theta$ and $r=4\sin\theta$ (Ans. $\frac{45\pi}{2}$)

(2) Evaluate $\iint \frac{r\,dr\,d\theta}{\sqrt{a^2+r^2}}$ over one loop of the curve $r^2=a^2\cos 2\theta$. (Ans. $\frac{a(4-\pi)}{2}$)

(3) Show that $\iint_R r^2 \sin\theta\, dr\, d\theta = \frac{2a^3}{3}$, where R is the semicircle $r=2a\cos\theta$ above the initial line.

(4) Find, using a double integral, the area of the cardoid. $r=a(1+\cos\theta)$ (Ans. $\frac{3\pi a^2}{2}$)

(5) Using double integration, find the area enclosed by the curves.

(a) $y=2x$ and $y^3=2x$ lying in the first quardant.

(b) $x=y-y^2$ and $x+y=0$

(c) $y^2=4-x$ & $y^2=4-4x$

(Ans. a) $\frac{1}{8}$ b) $4/3$ c) 8)

(6) Find the area (using double integration)

(a) of the curve $r^2=a^2\cos 2\theta$ (Ans. a^2)

(b) situated outside the circle $r=2a\cos\theta$ and inside the cardioid $r=a(1+\cos\theta)$ (Ans. $\frac{\pi a^2}{2}$)

(c) situated inside the circle $r=2a\cos\theta$ and outside the circle $r=a$ (Ans. $a^2\left(\frac{\pi}{3}+\frac{\sqrt{3}}{2}\right)$)

(7) Find the mass of a plate in the form of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, the thickness at any point varying as the product of its distances from the axes. (Ans. $\frac{Ka^2b^2}{2}$)

(8) Find the volume bounded by the cylinder $x^2+y^2=4$ and the planes $y+z=3$, and $z=0$ (Ans. 12π)

(9) Find the volume of the portion of the cylinder $x^2+y^2=1$ intercepted between the plane $x=0$ and the paraboloid $x^2+y^2=4-z$ (Ans. $\frac{7\pi}{2}$)

(10) Verify Green's theorem in the plane for

$$\int_C (x^3-x^2y)\,dx + xy^2\,dy$$

where C is the boundary of the region enclosed by the circles $x^2+y^2=4$ & $x^2+y^2=16$ (Ans. 120π)

(11) Using Green's theorem in a plane, evaluate the line integral $\int_C x^2(1+y)\,dx + (y^3+x^3)\,dy$ (Ans. $\frac{3}{2}$) where C is the square bounded by $y=\pm 1$ & $x=\pm 1$

(12) Use Green's theorem to find the value of

$$\int_C (xy-x^2)\,dx + (x^2y)\,dy$$

(Ans. $\frac{-1}{12}$)

along the closed curve C formed by $y=0$, $x=1$ & $y=x$.

(13) Evaluate $\int_C (2xy-x^2)\,dx + (x+y^2)\,dy$ where C is the closed curve of the region bounded by $y=x^2$ and $y^2=x$ (Ans. $\frac{1}{30}$)

(14) Evaluate by using Green's theorem,

$\int_C x^2y\,dx + y^3\,dy$ where C is the closed path formed by $y=x$ & $y^3=x^2$ from $(0,0)$ to $(1,1)$

(Ans. $\frac{-1}{44}$)

(15) Find the volume of a sphere of radius a (Ans. $\frac{4}{3}\pi a^3$)

(16) Prove by using a double integral, that the volume generated by the revolution of the cardioid $r = a(1+\cos\theta)$ about its axis is $\frac{8}{3}\pi a^3$.

(17) Find the volume of the tetrahedron bounded by the co-ordinate planes and the ~~sphere~~ plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

(Ans. $\frac{1}{6}abc$)

(18) Evaluate $\iiint \frac{1}{x^2+y^2+z^2}\, dx\,dy\,dz$ through out the volume of the sphere $x^2+y^2+z^2 = a^2$. (Ans. $4\pi a$)

(19) Show by Applying Green's theorem that the area bounded by a simple closed curve C is $= \frac{1}{2}\int x\,dy - y\,dx$ and hence find the area of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

(Ans. πab)

(20) Verify Green's theorem in the plane for $\int_C (x^3 - x^2y)\,dx + xy^2\,dy$ where C is the boundary of the region enclosed by the circles $x^2+y^2=4$ and $x^2+y^2=16$.

(Ans. 120π)

(21) Verify Stoke's theorem for
$F = (x^2+y^2-4)\hat{i} + 3xy\,\hat{j} + (2xz+z^2)K$
where S is the surface of the hemisphere $x^2+y^2+z^2=16$ above the xy-plane & C is its boundary. (Ans 0)

(22) Verify Stoke's theorem for the function $F = y^2\hat{i} + xy\hat{j} - xzk$ over the upper half of the sphere $x^2+y^2+z^2=a^2$ (Ans. 0)

(23) Verify Stoke's theorem for $F = (y-z+2)\hat{i} + (yz+4)\hat{j} - xzK$ where S is the surface of the cube $x=0, y=0, z=0, x=2, y=2, z=2$ above xy-plane. (Ans. -4)

(24) Verify Stoke's theorem for a vector field defined by
$F = (x^2-y^2)\hat{i} + 2xy\,\hat{j}$ (Ans. $2ab^2$)
in the rectangular region in the XOY plane bounded by the lines $x=0, x=a, y=0$ and $y=b$.

(25) Verify the divergence theorem for $F = 4x\hat{i} - 2y^2\hat{j} + z^2k$ taken over the region bounded by $x^2+y^2=4, z=0$, and $z=3$ (Ans. 84π)

(26) Evaluate the surface integral $\iint_S A\cdot\hat{n}\,ds$.
if $A = yz\hat{i} + 2y^2\hat{j} + xz^2\hat{k}$ and S is the surface of the cylinder $x^2+y^2=9$ contained in the first octant between the planes $z=0$ and $z=2$. (Ans. 108)

(27) Verify the divergence theorem for the function
$F = y\hat{i} + x\hat{j} + z^2\hat{k}$ over the cylindrical region bounded by $x^2+y^2=a^2, z=0, z=k$. (Ans. $k^2a^2\pi$)

(28) Verify the divergence theorem for the function
$F = 2xz\hat{i} + yz\hat{j} + z^2k$ over the upper half of the sphere $x^2+y^2+z^2=a^2$
(Ans. $\frac{5\pi a^4}{4}$)